Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Describe how a fixed loop is used in the program.

- for b0 = 1 to 20 … next b0 surrounds the sequence
- After 20 repeats the program moves on to end

3 Quick check

- a) Diamond
- b) Faster
- c) Fixed loop
- d) Fewer components, so more reliable

4 Key terms

- 1 – D
- 2 – F
- 3 – B
- 4 – A
- 5 – C
- 6 – E

5 Draw the flowchart symbol for a decision and for an output.

- Decision: diamond with yes/no branches
- Output: parallelogram (or rectangle with "high/low")

6 Describe how the program could switch on a lamp when the reading is below 80.

- readadc 1, b1 to read the value into a variable
- if b1 < 80 then high 4 (else low 4)

7 Write a program for the flowchart.

- main: label and if pin0 = 1 then (or input check loop)
- high 2
- pause 3000
- low 2 and goto main